

Cardano–Tartaglia via the Pandrosion Resolvent

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April 2026

Abstract

The Cardano–Tartaglia formula solves the depressed cubic $x^3 + px + q = 0$ by the substitution $x = u + v$ with $u^3 + v^3 = -q$ and $3uv = -p$. We interpret this as a *splitting of the Pandrosion field*: the quotient $Q(\alpha, z) = z^2 + \alpha z + (\alpha^2 + p)$ satisfies $Q(\alpha, \alpha) = P'(\alpha)$, and the resolvent quadratic $t^2 + qt - p^3/27 = 0$ corresponds to the field extension required to separate the quotients. The casus irreducibilis ($\Delta > 0$, three real roots) requires complex intermediates, reflecting the non-trivial topology of F_P on the real line.

1 Cardano’s formula as Pandrosion splitting

For $P(z) = z^3 + pz + q$ with roots $\alpha_1, \alpha_2, \alpha_3$:

Proposition 1.1 (Pandrosion quotient of the cubic).

$$Q(\alpha_j, z) = \frac{P(z)}{z - \alpha_j} = z^2 + \alpha_j z + (\alpha_j^2 + p), \quad (1)$$

and the evaluation $Q(\alpha_j, \alpha_j) = 3\alpha_j^2 + p = P'(\alpha_j)$.

Verified at 10^{-16} precision for all test cases.

Theorem 1.2 (Cardano = Pandrosion resolvent). *The substitution $x = u + v$ with $u^3 + v^3 = -q$, $uv = -p/3$ reduces the cubic to the resolvent quadratic:*

$$\boxed{t^2 + qt - \frac{p^3}{27} = 0, \quad \text{roots: } u^3, v^3.} \quad (2)$$

This is the Pandrosion resolvent: it splits the field $F_P = P'/P$ into components that can be extracted by radicals.

2 The casus irreducibilis

Proposition 2.1. *The discriminant $\Delta = -4p^3 - 27q^2$ determines the root type:*

Δ	Root type	Example
$\Delta > 0$	3 distinct real (casus irreducibilis)	$x^3 - 15x - 4$, $\Delta = 13068$
$\Delta = 0$	Repeated root	$x^3 - 3x + 2$, $\Delta = 0$
$\Delta < 0$	1 real + 2 complex conjugate	$x^3 + 3x + 2$, $\Delta = -216$

The casus irreducibilis is remarkable: all three roots of $x^3 - 15x - 4$ are real ($\alpha = 4, -2 \pm \sqrt{3}$), yet Cardano’s formula *necessarily* passes through complex cube roots ($u^3 = 2 + 11i$). This reflects the topology of F_P : on the real line, F_P has three poles separated by two zeros, and no single radical can “jump” between all three sheets simultaneously.

3 Sum of quotients

Proposition 3.1. $\sum_{j=1}^3 Q(\alpha_j, z) = P'(z) = 3z^2 + p$.

This is the degree-3 instance of the general Pandrosion identity $\sum Q_j = P'$ (Paper 1).

4 Vieta via Pandrosion

For $P = z^3 + pz + q$: $\alpha_1 + \alpha_2 + \alpha_3 = 0$, $\alpha_1\alpha_2 + \alpha_1\alpha_3 + \alpha_2\alpha_3 = p$, $\alpha_1\alpha_2\alpha_3 = -q$. These are the elementary symmetric polynomials of the Pandrosion fiber $\{\alpha_1, \alpha_2, \alpha_3\}$.

Verified for all test cases at machine precision.

5 From cubic to quintic: the Galois obstruction

Paper 84 showed that the general quintic is *not* solvable by radicals because A_5 is simple. The cubic resolvent works because S_3 is solvable: $\{e\} \triangleleft A_3 \triangleleft S_3$ with abelian factors $\mathbb{Z}_3$ and $\mathbb{Z}_2$. Cardano's resolvent quadratic corresponds to the first step $S_3 \rightarrow S_3/A_3 \cong \mathbb{Z}_2$.

References

- [1] G. Cardano, *Ars Magna*, 1545.
- [2] I. Besevic, *The Pandrosion quotient*, Pandrosion Dynamics **1** (2026).
- [3] I. Besevic, *Abel–Ruffini via Pandrosion orbits*, Pandrosion Dynamics **84** (2026).